MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 7 of 10 (Start time: 02:12:39 PM, 21 June 2025)

Integration of 3x^2 with respect to x is.....

Select the correct option

○	x^4
○	x
○	x^2
◉	x^3

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 5 of 10 (Start time: 02:12:11 PM, 21 June 2025)

lnx^2 can be written as _____.

Select the correct option

○	lnx/2
○	2+lnx
○	2-lnx
◉	2lnx

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 1 of 10 (Start time: 02:05:39 PM, 21 June 2025)

The problem of finding the rate at which one quantity changes w.r.t another quantity is an example of related rates.

Select the correct option

○	False
◉	True

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Question # 2 of 10 (Start time: 02:11:26 PM, 21 June 2025)

(1^3)+(2^3)+(3^3)+....+(20^3) equals ---------

NOTE: x^n means 'x' to the power 'n'

Select the correct option

●	44100
○	42925
○	None of these
○	34548

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 10 of 10 (Start time: 02:08:25 PM, 21 June 2025)

$1^2+2^2+3^2+_+15^2$ equals _____.

Select the correct option

○	15(16)(17) / 6
○	15(16)(29) / 6
○	15(16)(30) / 6
●	15(16)(31) / 6

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 9 of 10 (Start time: 02:13:17 PM, 21 June 2025)

What is the estimated area under $f(x) = 9 - x^2$ from $x = 0$ to $x = 4$ with mid points for $n = 1$?

Select the correct option

○	21
○	20
●	28
○	24

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 2 of 10 (Start time: 02:05:55 PM, 21 June 2025)

1+2+3+...+539 equals _____.

Select the correct option

○	538(540)/2
◉	539(540)/2
○	538(539)/6
○	538(540)/6

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 9 of 10 (Start time: 02:08:11 PM, 21 June 2025)

What the derivative of e^x ?

Select the correct option

○	e^{-x}
●	e^x
○	$-e-^x$
○	$-e^x$

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 1 of 10 (Start time: 02:11:06 PM, 21 June 2025)

$\frac{d}{dx}(\ln u) =$ _____ .

Select the correct option

○	$\frac{du}{dx}$
○	$x \ln x - x + C$
○	$\frac{1}{u}$
●	$\frac{1}{u} \cdot \frac{du}{dx}$

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 8 of 10 (Start time: 02:13:04 PM, 21 June 2025)

In approximation to an area Rn (where n is subscript) when limit is taken as n goes to infinity, approximation becomes actual area.

Select the correct option

- ○ False
- ◉ True

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MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 3 of 10 (Start time: 02:11:40 PM, 21 June 2025)

$\log_b M + \log_b N =$ _____.

Select the correct option

- ○ $\log_b \frac{M}{N}$
- ○ $\log_b M - \log_b N$
- ◉ $\log_b M\,N$
- ○ $M \log_b N$

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 5 of 10 (Start time: 02:06:41 PM, 21 June 2025)

The integral $\int \cos(5x)\,dx$ will be equal to ?

Select the correct option

○	$5\sin(5x)+c$
○	$-\frac{\sin(5x)}{5}+c$
◉	$\frac{\sin(5x)}{5}+c$
○	None of these

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 4 of 10 (Start time: 02:11:58 PM, 21 June 2025)

Let f be defined on an interval, and x_1, x_2 are the points in that interval. Then f is decreasing on the interval if $f(x_1) > f(x_2)$ whenever

Select the correct option

○	$x_1 = x_2$
◉	$x_1 < x_2$
○	$x_1 > x_2$
○	$x_1 \geqslant x_2$

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 6 of 10 (Start time: 02:12:24 PM, 21 June 2025)

To get better approximation to actual area under a continuous curve over a closed interval, we have to increase .....

Select the correct option

○	Width of the subintervals
○	Total area
◉	Number of subintervals
○	Size of the interval

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 8 of 10 (Start time: 02:07:58 PM, 21 June 2025)

The function f is said to be concave down on (a, b), if_____ on that interval.

Select the correct option

◉	$f''(x) < 0$
○	$f''(x) > 0$
○	$f''(x) \geqslant 0$
○	$f''(x) = 0$

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MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 3 of 10 (Start time: 02:06:14 PM, 21 June 2025)

Let f(x)=x^3 and a= -5 ,b=5 and n= no. of partition of [-3,3]=4
Let the widths of first, second, third and fourth intervals are 3,0,1,2 respectively. Then the "Mesh size" of partition is.....

Select the correct option

◉	3
○	0
○	1
○	2

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 4 of 10 (Start time: 02:06:26 PM, 21 June 2025)

What does the indefinite integral $\int f(x)dx$ represent?

Select the correct option

○	Curvature of the curve
◉	Families of antiderivative of the function $f(x)$
○	Area under the curve
○	None of these

MTH101 - Calculus And Analytical Geometry (Quiz No. 3) Quiz S

Question # 10 of 10 (Start time: 02:13:29 PM, 21 June 2025)

For any continuous function on the interval [0,1], if the area under this curve is divided into '5' equal rectangles ,then the length of each rectangle will be.....

Select the correct option

◉	1/5
○	1/2
○	1/4
○	1

Click to Save A

MTH101 - Calculus And Analytical Geometry (Quiz No. 3)

Question # 6 of 10 (Start time: 02:07:23 PM, 21 June 2025)

If $y = f(t)$ represent a position of an object at time t and $f'(t)$ represent the velocity of the object then what does $|f'(t)|$ represent?

Select the correct option

- ○ Direction of the object
- ◉ Speed of the object
- ○ Acceleration of the object
- ○ Velocity of the object

Click

MTH101 - Calculus And Analytical Geometry (Quiz No. 3) **Quiz Start Time: 02:05 PM, 21 June 2025**

Question # 7 of 10 (Start time: 02:07:41 PM, 21 June 2025) **Total Marks:** 1

In Rolle's theorem, f(x) is continuous in closed interval [a,b] and differentiable in open interval (a,b).Then why we don't discuss its differentiability in the closed interval [a,b] because ----------

Select the correct option

- ◉ f '(a) and f '(b) never exist
- ○ f(a) and f(b) never exist
- ○ f(a)=f(b)=0
- ○ f '(a)=f '(b)=0

Click to Save Answer & Move to Next Question